PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 – enacted by the Govt. of India)

SYLLABI

Course Outline for

B.Voc (Food Processing and Preservation)

as per NEP 2020

w.e.f session 2025-26

Course Outline for B.Voc (Food Processing and Preservation) as per NEP 2020 w.e.f. session 2025-26

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BFP 101	Core Discipline	Fundamentals of Food Science	2	2	4	40	10	40	10	100
BFP 102	Core Discipline	Bakery and Confectionery Technology	2	2	4	40	10	40	10	100
BFP 103	Core Discipline	Quality Assurance	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BFP 201	Core Discipline	Food Preservation Technology	2	2	4	40	10	40	10	100
BFP 202	Core Discipline	Food Plant Layout	2	2	4	40	10	40	10	100
BFP 203	Core Discipline	Dairy Science and Technology	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - Bakery Technician

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in Food Processing and Preservation = 48 + 4 #.

#Exit 1: Certificate Course - Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BFP 301	Core Discipline	Nutrition and Food Chemistry	2	2	4	40	10	40	10	100
BFP 302	Core Discipline	General Microbiology	2	2	4	40	10	40	10	100
BFP 303	Core Discipline	Foods Of Plant Origin	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC III	Skill Enhancement Course	SEC Options[1]			3					75
B.Voc. MDSC 301	Multidisciplinary Course	Entrepreneurship Development[1]	3	-	3	60	15	-	-	75
VAC II	Value-Added Course	VAC Options[3]			2					50
PW II	Project Work	Project Work-II & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BFP 401	Core Discipline	Food Packaging Technology	2	2	4	40	10	40	10	100
BFP 402	Core Discipline	Food Microbiology	2	2	4	40	10	40	10	100
BFP 403	Core Discipline	Foods Of Animal Origin	2	2	4	40	10	40	10	100
BFP 404	Core Discipline	Documentation And Record Keeping	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
VAC III	Value-Added Course	VAC Options[3]			2					50
SIT 401	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - Food Microbiologist

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 5th Semester. Viva-Voce examination to be held within 3 weeks of the start of 5 th semester.

** Minimum Credits Required to get Diploma Course in Food Processing and Preservation = 96 + 4 #.

#Exit 2: Diploma Course - Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the second year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF B.VOC
(**Food Processing and Preservation**)
(SEMESTER 1 to 4)

PROGRAMME OUTCOMES

PO1. Understanding of the concepts, principles of working of the Food Industry.

PO2. To understand the different preservation techniques of foods by heat processing, by low temperature, by removal of water and by using salt and sugar or by other relevant methods._.

PO3. To identify the various microorganisms pathogenic or non-pathogenic and trace the cause of food contamination.

PO4. Describe the spoilage and deterioration mechanisms happening in foods and food products and to apply methods to control microbial activities prolonging shelf life, assuring it fit for human consumption.

PO5. To plan and design a Food Plant Layout in accordance with the type of establishment (small, medium or large) in a specific area.

PO6. To understand Food laws and Food Safety regulations in India and worldwide, this will be accompanied by Safety and hygienic practices to be followed in a food plant layout.

PO7. To develop ability to apply principles of food engineering in industry,

PO8. Process and prepare food products which could be brought in the concurrent food market for sale.

PO9. To develop awareness of the marketing strategies, management hierarchy involved in a food processing plant.

Semester I

<table>
<tr><td>1</td><td>Course Title</td><td>Fundamentals of Food Science</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 101</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Prac</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Bakery Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To understand the importance of nutrients in our daily diet and to formulate nutritionally enriched food products.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Develop proficiency skill in producing different nutritious food products. .
CO2. Make different processed food products with quality assurance. .
CO3.Understand the structure, composition, and nutritional quality of foods.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Introduction to Food Science: Definition & Scope. Basic concept of Food: Nutrient, Nutrition, Classification of Nutrients : Macronutrients, Micronutrients; dietary guidelines. Concept of food groups. **UNIT– II** Constituents of food - Definition, occurrence, properties, significance : Carbohydrates, Proteins, Fats and oils; Nutritive aspects. Protein quality. Bioavailability of nutrients. Vitamins. Minerals. Fiber. Water. **UNIT– III** Food and nutrition - Energy metabolism - BMR. Recommended dietary allowances, Balanced diet for different age groups (Infancy to old age). Stability of nutrients. Role of nutrients in health and disease prevention. **UNIT-IV** Food Processing: Unit operations in food industries : Materials handling, cleaning, separating, grinding, mixing, heat exchanging, forming, packaging. **Practical Work** 1. Demonstration of basic laboratory equipments used in food processing labs. 2. Demonstration of chemistry of candy making and the effect of temperature on the texture of candies. 3. Calculation of BMR 4. Enrichment and fortification of routine diet. 5. Demonstration of unit operation in food processing :mixing, separating, heat exchanging, packaging.
12	**Suggested Resources for Reading** 1. Food Science by B. Srilalshmi, New Age International Private Limited; Sixth edition, 2015. 2.Food: Facts and Principles by N. Shakuntala Manay, M. Shadaksharaswamy, New Age International Private Limited; Fourth edition, 2020. 3. Food Science by Joseph H. Hotchkiss and Norman N Potter,Aspen Publishers Inc.,U.S, Fifth edition,2007 4. Essentials of Food Science by Vickie A. Vaclavik, Elizabeth W. Christian, Springer Publications,Fourth edition,2013 5. Food Science by Sunetra Roday., Oxford publication, Third edition, 2018

<table>
<tr><th>1</th><th>Course Title</th><th>Bakery and Confectionary Technology</th></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 102</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Bakery technician.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To impart basic knowledge of scope of bakery and confectionery and learn about baking process.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Familiarizing the students with different equipment and ingredients used in bakery unit and their functions..
CO2. The students will be able to identify various bakery and confectionery products
CO3.The learners will develop fundamental baking techniques.
CO4. Work effectively as a member of a team and learn entrepreneurial skills for self reliance.
CO5. The students will learn the basics of setting up a bakery unit.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Introduction- Structure of wheat grain, composition and milling process, Ingredients used in bakery process and their role, types of flour, leavening agents and its types. **UNIT– II** Baking methods: Various Methods of Bread Making: (Straight dough method, Sponge dough method.) Various Cake making ,methods(Sugar batter, Flour batter method), Pastry Making: Principle and types **UNIT– III** Bakery technology- Sensory attributes of bakery products and factors affecting it, Bread faults and remedies, Various equipments used in bakery industry- Small and large equipments and their role **UNIT-IV** Confectionary- Types of Confectionery, raw materials and processing of toffee, Fondant, hard boiled sweets. Sugar Panned Confections and their types. **Practical work** 1. Preparation of White Bread. 2. Preparation of Fruit cake. 3. Studying the effect of temperature on the process of biscuit making. 4. Preparation of Garlic bread and sensory analysis. 5. Preparation of Icing of Cakes. 6. To estimate WAP of the flour sample 7. To determine gluten content in the dough
12	**Suggested Resources for Reading** 1. Textbook of bakery and confectionery, by Yogambal Ashok Kumar 2. Bakery products science and technology,by Weibiao zhou, Wiley blackwell, second edition 3. A professional Text to bakery and confectionary by John Kingslee 4. Bakery-I: Student handbook and practical manual- CBSE. 5. Bread: A baker's book of techniques and recipes by Jeffrey Hamelman-Wiley, 2012. 6.Confectionery Science and Technology by _Richard W. Hartel · Joachim H. von Elbe Randy Hofberger,Springer ,2017

<table>
<tr><th>1</th><th>Course Title</th><th>Quality Assurance</th></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 103</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Bakery Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To develop a broad understanding of the principles of modern quality management systems; to understand objective and subjective factors that contribute to the quality of foods or food-related services within the food value chain.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1.Illustrate the importance of food quality standards and of Quality Assurance.
CO2.Understand the principles of Total Quality Management to evaluate and select between alternative strategies in designing a quality management system.
CO3.Explain the application of food quality and food safety system.
CO4.Understand the role of food quality assurance system standards in India.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Objectives, importance and functions of quality control. Principles of food quality control and quality assurance, quality control and assessment in food materials used in the industry. **UNIT– II** Total quality management; Good Manufacturing Practices, Good Hygienic Practices, Good Laboratory Practices, Quality Circles, general awareness and role of management practices in quality control. **UNIT– III** Food laws and standards : Food regulations and standards in Indian scenario, concepts of Codex Alimentarius, HACCP, USFDA, ISO 9000 series etc. Food standards and safety Act: salient provisions and prospects. **UNIT-IV** Sensory quality evaluation: Objectives, type of food panels, characteristics of panel member, layout of sensory evaluation laboratory, Types of Sensory Testing : difference tests, Rating tests, Sensitivity test, Descriptive test. Chemical dimension of basic tastes. **Practical work** 1.Techniques of quality assessment of different natural and processed foods. 2 Identification and ranking of food product attributes 3. Sensory methods for measuring food quality assessment of raw materials used and processed bakery products 4. Study of cleaning and sanitizers used in pre and post-operative processes in bakery industry 5. Documentation of details of baking ingredients, process and finished products used in baking industry
12	**Suggested Resources for Reading** 1. Food Quality Assurance Principles and Practices by InteazAlli, CRC Press, 2004. 2. Quality Assurance for the Food Industry. A Practical Approach by J Andres Vasconcellos , CRC Press, 2003 3. Safety And Quality Assurance In Food Supply Chain: Emerging Technologies and challenges by K.P. Sudheer, Bindu Lakshmanan New India Publishing Agency (2021) 4. Food Science by Sunetra Roday., Oxford publication, Third edition, 2018 5. Sensory Evaluation Techniques by Morten C. Meilgaard, B. Thomas Carr, Gail Vance Civille ,CRC Press LLC, Fourth edition, 2006. 6. Principles of Sensory Evaluation of food by Amerine, Pangborn & Roessler , Academic Press, London.2013

Semester 2

<table>
<tr><td>1</td><td>Course Title</td><td>Food Preservation Technology</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 201</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Bakery Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>The goal of this course is to provide students with a fundamental understanding of food preservation</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Will be able to comprehend the reason for food spoilage.
CO2. Will be able to comprehend the principles of preventing contamination and the removal of microorganisms methods used for food safety.
CO3. Students will be able to understand various heat preservation techniques, low temperature techniques.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** Introduction: scope and benefit of industrial food preservation, perishable, non-perishable food, causes of food spoilage. Preservation by salt and sugar – Principle, method, equipment and effect on food quality **UNIT– II** Thermal processing methods of preservation – Canning, blanching, pasteurization, sterilization, evaporation, Preservation by fermentation- Definition, advantages, disadvantages, types, equipments **UNIT– III** Food preservation by use of low temperature – Principle, processes and effect on quality. Preservation by drying, dehydration and concentration – Principle, Methods, equipment and effect on quality of food. **UNIT-IV** Recent methods in preservation: Pulsed electric field processing, High pressure processing, Processing using ultrasound,Microwave heating, Ohmic and infrared heating . **Practical work** 1. Demonstration of various machineries used in food processing. 2. Demonstration on the effect of blanching on the quality of foods. 3. Preservation of food by high concentration of sugar i.e. preparation of jam 4. Preservation of food by using salt 5. Preservation of milk by condensation/concentration 6. To Demonstrate the Preservation of Fruits by drying
12	**Suggested Resources for Reading** 1. Food Science- Potter, CBS publishers. 2. Food Processing and Preservation- Subbulaksmi G., and Udipi S 3. Principles of Food Science, Vol. II- G. Borgstron, Mc. Millan Co. Ltd. London 4. _Principles of food preservation Part I& II- Owen R. Fenemma. 5. Technology of Food Preservation - N.W. Desroiser and N.W. Desrosier

<table>
<tr><th>1</th><th>Course Title</th><th>Food Plant Layout</th></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 202</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Bakery Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>• Exposure of the students to the basic setup of a Food industry
• To make them conversant with the machinery and equipment in a food industry.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. understand concepts of plant layout.
CO2. have knowledge on building, utilities in the plant.
CO3. know the importance of proper food plant design and safety.
CO4. evaluate and develop problem solving skills regarding food processing operations that can affect the quality of foods.
CO5. know and implement good manufacturing practices.
.....</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** **UNIT– I** **Industrial plant layout and design:** Basic concepts of Plant Layout with reference to food industries. Hygiene requirements in food plant layout and design. **UNIT– II** **Plant Design**: Design consideration for location of food plants. Basic understanding of equipment layout and ventilation in food processing plants. Preparation of flow sheets for material movement and utility consumption in food plants. **UNIT– III** **Plant Layout** : Preparation of a Plant Layout, Plant Layout problem, importance, objectives, classical types of layouts. Evaluation of layout. Advantages of good layout **UNIT-IV** **Plant Layout Requirements :** Common materials of construction of Food plant, building. Maintenance of Food Plant Building, Illumination and ventilation, Cleaning and sanitization, painting and color coding, Fly and insect control. **Practical Work :** Industrial visit and report making.
12	**Suggested Resources for Reading** 1. Hygienic Design of Food Factories, by John Holah, H. L. M. Lelieveld, Elsevier Publication, 2011 2.Practical Design, Construction and Operation of Food Facilities by J. Peter Clark, Academic Press Publishers.2008 3. Food Plant Economics by Zacharias B. Maroulis, George D. Saravacos ,CRC Press Publishers.2007 4., Food Plant Design by Antonio Lopez-Gomez, Gustavo V. Barbosa-Canovas, CRC Press Publishers.First edition, 2005

1	**Course Title**	**Dairy Science and Technology**
2	**Course Code**	BFP 203
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> **(4 X 2 marks) = 8 marks** **(4 X 8 marks) = 32 marks** i. The exam of this course will be of 3 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months be **iv.** Use of non-programmable calculators by the students in the Examination Hall is allowed. The cal provided by the University/College to the examinees**.** **Instructions for paper setters and candidates:** i. The syllabus of this paper has been divided into FOUR units. ii. All questions carry equal marks iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Q (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students a out of 6 short questions of 2 marks each. iv. The students are required to attempt one long question of 8 marks from each unit.
7	**Job Role**	**Bakery Technician**
8	**Course Objectives**	Students will learn about _Principles of Dairy, processing of technologies to handle physical aspects of fluid based milk and its various products.
9	**Course Outcomes**	On successful completion of the course, the Students will be able to CO1. In-depth knowledge of dairy production and processing techniques. CO2.Comprehensive understanding of dairy production, processing, and preservation techniques. CO3.Better understanding of the concepts underlying Dairy processing as well as the reliability and caliber of Dairy products. CO4. Students will learn about technologies to handle physical aspects of fluid based process CO5.Students will learn about various equipments present in advanced dairy
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

11	**Course Content** UNIT– I Introduction- Milk definition, composition of milk, nutritive value of milk, physico-chemical properties of milk constituents and physico- chemical properties of milk, factors affecting quality of milk. UNIT– II Market Milk- Manufacture, packaging and storage of standardized pasteurized milk, Various methods of pasteurization, Judging and grading of milk, Flavour defects in milk, their causes and Prevention, UNIT– III Special milks- Sterilized milk, Homogenized milk, Flavoured milks, Reconstituted/Rehydrated milk, Recombined milk, Toned milk, Double Toned milk Fermented milk- Acidophilus milk, bulgarian buttermilk, Kumiss and kefir. UNIT-IV Dairy Products - Composition and manufacturing of butter, butteroil, Ice cream and cheese **Practical work** 1. To perform platform tests for milk analysis- analysis of milk for its organoleptic properties 2. To determine stability of milk using COB Test 3. To test milk quality by performing MBRT Test 4. To perform Alcohol test to assess raw milk quality 5. Sediment test to check for adulteration in milk sample 6. Determination of titratable acidity of milk 7. To determine Fat content by Gerber method
12	**Suggested Resources for Reading** 1.Outlines of Dairy Technology, by Sukumar De, Oxford University Press. 2. Milk Hygiene In Milk Production Processing and Distribution, F.A.O Publication. Note: Latest edition of text books may be used.

Semester III

1	**Course Title**	**Nutrition and Food Chemistry**
2	**Course Code**	BFP 301
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
5	**Credits**	4 Credits

<table>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To understand the importance of nutrients in our daily diet and to formulate nutritionally enriched food products.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Describe details of the physical and chemical interactions between food components and their impact on quality.
CO2.Understand the structure, composition, and nutritional quality of foods.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** UNIT– I Carbohydrates- Structure and Classification of important carbohydrates, glycoproteins and glycolipids. Food sources of carbohydrates. Dietary fibre. UNIT– II Proteins- Structure and classification of amino acids, Food sources of proteins. Enzymes- Classification & Nomenclature. Coenzymes, Factors affecting enzyme efficiency. Enzyme kinetics. UNIT– III Lipids- Structure & Classification, saturated, unsaturated and phospholipids. Physical properties- melting point, smoke, flash and fire point. Chemical properties- reichert meissel value, polenske value, iodine value, peroxide value, saponification value. UNIT-IV Changes in food during processing- Protein denaturation, Oxidative and Hydrolytic Rancidity. Concept of Gelatinisation and retrogradation of starches, Effect of frying on fats. Practical Work 1. Estimation of minerals. 2. Determination of thermal inactivation time of enzymes in fruits and vegetables. 3. Estimation of iodine value. 4. Estimation of peroxide value. 5. Protein estimation in food items. 6. Carbohydrate estimation in food items.
12	**Suggested Resources for Reading** 1. Aurand, L. W. and Wood, A. E. (1973). Food Chemistry. The AVI Publishing Co., Connecticut. 2. Belitz, H. D., Grosch, W. and Schieberler, P. (2004). Food Chemistry. Springer, Berlin. 3. DeMan, J. M. (1999). Principles of Food Chemistry. A Chapman and Hall Food Science Book, Aspen Publ., Inc., Gaithersburg, Maryland. 4. Fennama, O. R. (ed). (1996). Food Chemistry. Marcel Dekker, Inc., New York.

<table>
<tr><td>1</td><td>Course</td><td>General Microbiology</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 302</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To understand microorganisms and the concept of cleanliness, sterilization, maintenance of lab equipment, microbial growth, microscopy and culture medium.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Gain knowledge about the different cell organelles of microorganisms and their detailed functions.
CO2. Study the growth and control of microbes
CO3. Learn about the different types of microorganisms and their significance.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** UNIT– I History of Development of Microbiology- Spontaneous generation vs. biogenesis. Contributions of Anton von Leeuwenhoek, Louis Pasteur, Robert Koch, Joseph Lister, Alexander Fleming. Control of microorganisms: Concept of sterilization and pasteurization, physical and chemical agents to control microbial UNIT– II Microbial growth- phases of growth, growth curve, generation time, specific growth rate, optimum growth. Chemostat and turbidostat. Concept of culture medium: Types based on: consistency, constituents/ ingredients UNIT– III Light microscope-Bright Field Microscopy, Dark Field Microscopy, Fluorescent Microscopy, Electron Microscope- Transmission Electron Microscopy, Scanning Electron Microscopy UNIT-IV Characteristics of Predominant microbes in food- Bacteria, yeasts, molds characteristics and structure, Structure of prokaryotic cells, cell wall of Gram Positive and Gram negative bacteria with structure. Practical work 1. Study of basic equipment: laminar flow, homogenizer, autoclave, biosafety systems 2. Sterilization of glassware and other lab equipment 3. Preparation and sterilization of liquid and solid media for isolation of microbes 4. Inoculation and other aseptic procedure 5. To study the parts and working of a compound microscope.
12	**Suggested Resources for Reading** 1. Pelczar MJ, Chan ECS and Krieg NR. Microbiology. McGraw Hill Book Company. 2. Stanier RY, Ingraham JL, Wheelis ML, and Painter PR. General Microbiology. McMillan. 3. Tortora GJ, Funke BR and Case CL. Microbiology: An Introduction. Pearson Education

<table>
<tr><td>1</td><td>Course Title</td><td>Foods Of Plant Origin</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 303</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To develop a broad understanding of the principles of foods of plant origin and its classification</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to-

CO1. learn about various types of cereals, pulses and its types
CO2. know about fruits and vegetables, fats and oils, structure and types.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** UNIT– I Plant based foods: Nutritional composition of plant based foods and their benefits. Plant-based foods vs. animal-based foods. Introduction to plant-based diets: vegetarianism, veganism, and flexitarianism Cereals - Structure and composition of cereals, Malting, gelatinization of starch, types of cereals. Rice- Structure and composition, parboiling of rice. UNIT– II Pulses- Structure and composition of pulses, toxic constituents in pulses. Processing of pulses- soaking, germination, decortications, cooking and fermentation UNIT– III Fats and Oils- Classification of lipids, types of fatty acids - saturated fatty acids, unsaturated fatty acids, essential fatty acids, trans fatty acids. Refining of oils - steam refining, alkali refining, bleaching, steam deodorization, hydrogenation. UNIT-IV Fruits and Vegetables- Classification of fruits and vegetables, general composition, enzymatic browning, names and sources of pigments. Practical work 1. To study gelatinization behavior of various starches 2. To study the concept of gluten formation of various flours. 3. To study malting and germination. 4. To study dextrinization in foods.
12	**Suggested Resources for Reading** 1. Food Science by Sunetra Roday., Oxford publication, Third edition, 2018

Semester IV

1	**Course Title**	**Food Packaging Technology**
2	**Course Code**	BFP 401
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
5	**Credits**	4 Credits

<table>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To enable the students to understand about packaging and packaging materials, interaction of food items with packaging materials</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Comprehend the reason for food packaging.
CO2. Learn about different kinds of packaging
CO3. Understand about canning operations.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** Unit I Introduction to Food Packaging: Functions of packaging, Effect of environmental factors on quality of food. Estimation of shelf life, analysis of storage requirement, accelerated storage studies: Vacuum and Inert Gas Packaging: Tests on packaging materials, Mechanical strength (Tension, notch and tearing strengths), Gas and water vapor transmission rates. Unit II Metal Cans as Packaging material: Types of Metallic Cans: Tin cans and Aluminum cans. Specialty of Open top sanitary cans, Lacquers and their use, Three piece cans and two piece cans, Aerosol Cans. Introduction to Canning operations: Can Reformer, Flanger, Seaming, Can closures, Glass jars and Bottles in food packaging, Design features and applications, Sterilization of jars and bottles. Unit III Flexible Films Packaging: Formation of Films and pouches, Plastics used and their Specific applications.Rigid and Semi rigid plastic packaging methods. Copolymers and their applications.Co-extruded films and Laminates. Extrusion –Retort pouch packaging. Laminated Paper board Cartons, Fiber Board and Corrugated Card board packaging and their applications. Unit IV Filling And Sealing Operations For Various Types of Packages: Closing and sealing of rigid plastic containers. Filling and sealing of flexible plastic containers, Seal types, hot wire sealing, hot bar sealing and impulse sealing. Active packaging, Moisture control, CO_2 and Oxygen scavenging, Modified atmosphere packaging: principles and applications. Permeability of gases in packs. Practical work 1. To study quality and strength of packaging materials. 2. Measurement of cartons' dimensions as per organizational standards. 3. Measurement of dimensions of bottle mouth, cans and their caps. 4. Determination of tensile strength of given material. 5. Testing of chemical resistance of packaging materials. 6. Demonstration of sealing processes used in food industry. 7. Demonstration of filling process.
12	**Suggested Resources for Reading** 1. Coles, R., Dowell, D.M. & Kirwan, J, Food Packaging Technology, Black Well Publishing Ltd. 2. NIIR Board, Food Packaging Technology Handbook, National Institute of Industrial Research. 3. Pirenger, O.G. and Baver, A.L., Plastic Packaging Materials for Food, Wiley.

<table>
<tr><td>1</td><td>Course Title</td><td>Food Microbiology</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 402</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Students will study about the growth of different types of microorganisms based on various environmental factors and effect on food.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Learn about the different types of fermentation processes, equipments used and microbiological processes involved
CO2. Gain knowledge of significance and activities of microorganisms in food.
CO3. Know the microbial quality control and quality schemes used in food industries.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** Unit-I History and Development of Food Microbiology, Characteristics of Predominant microbes in food: Bacteria, yeasts, Moulds. Normal microbiological quality of foods and its significance. Sources of microbes in food: Soil, air, water, plants, animals, food ingredients, equipment. Structure of prokaryotic cells, cell wall of Gram Positive and Gram negative bacteria. UNIT– II Control of microorganisms: Concept of sterilization and pasteurization, physical and chemicalagents, mechanical removal methods. UNIT III Types of Fermentation process: Batch, continuous and fed batch and their significance. Surface and submerged fermentation. Basic concept of fermenters: parts and their function. Fermented Food: Vegetables, dairy products, meat, poultry and fish products. Cocoa, coffee, wine, beer and their health benefits. Concept and application of prebiotics and probiotics. Unit IV Microbial Food Spoilage: Factors responsible for spoilage, spoilage of specific food types (Milk and dairy products, Meat, poultry and seafoods, Cereal and cereal products, Fruits and vegetables products.) Spoilage bacteria in raw and pasteurized foods, canned foods, soft drinks and fruit juices, refrigerated foods. Indicators of microbial food spoilage. Practical Work : 1. Isolation of bacteria and fungus from spoiled food. 2. Identification of isolated bacteria by simple, gram staining and negative staining 3. Identification of fungi by lactophenol cotton blue staining. 4. Methylene Blue reductase test.
12	**Suggested Resources for Reading** 1.Frazier WC and Westhoff D C and Vanitha NM (2017): Food Microbiology, 5th Ed. MaGraw Hill Education.. 2. Jay JM (2005): Modern Food Microbiology, 3rd Ed. CBS Publishers & Distributors. 3.Pelczar M, Chan ECS, Krieg N(2009): Microbiology : Application Based Approach, Tata McGraw Hill Education.

<table>
<tr><td>1</td><td>Course Title</td><td>Foods of Animal Origin</td></tr>
<tr><td>2</td><td>Course Code</td><td>BFP 403</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Students will learn about processing of food from animal to handle its aspects of perishable nature.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Gain In-depth knowledge of food from animal sources.
CO2. Learn about production and processing techniques.
CO3. Understand of the concepts underlying Dairy processing as well as meat and poultry.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** UNIT– I Nutritional importance of animal origin food, various categories of food from animals. Dairy- different dairy animals, UHT Milk, yogurt processing UNIT– II Meat - Definition of carcass, concept of red meat and white meat, composition of meat, marbling, sources of meat, nutritive value of meats UNIT– III Poultry - Structure of hen's egg, composition and nutritive value, egg proteins, characteristics of fresh egg, deterioration of egg quality, difference between broiler and layers. UNIT-IV Fish - Classification of fish (freshwater and marine), aquaculture , composition of fish, characteristics of fresh fish, spoilage of fish- microbiological, physiological, biochemical. Practical work 1. Prepare a report on any food obtained from animals in detail from procurement till processing. 2. Determination of egg components (shell strength, egg weight and yolk index). 3. Grading and quality evaluation of egg.
12	**Suggested Resources for Reading** 1.Bawa. A.S, O.P Chauhan et al. Food Science. New India Publishing agency, 2013 2.Roday,S. Food Science, Oxford publication, 2011. Note: Latest edition of text books may be used.

1	**Course Title**	Documentation And Record Keeping
2	**Course Code**	BFP 404
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
5	**Credits**	4 Credits

<table>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4 X 2 marks) = 8 marks
(4 X 8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To enable the students to understand the importance of documentation in food industry and study preparation of records.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand documentation and understand preparation of records.
CO2. Prepare different types of documents as per requirement.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

11	**Course Content** UNIT– I Introduction- Documents and records: Concept of Documents and records, General principles for document and record development and maintenance. Importance of documents and record keeping in food industry. UNIT– II Document and record control: Procedures for monitoring, assessing and verification of records. Deviation procedures in documents and records. Procedures for deviation correction. Filing and storing systems for records in industry- manual and electronic UNIT– III Record keeping and hazard control: HACCP system, Types of HACCP records, Control of documents, Retention of records, Review of records, disposal and retrieval of HACCP records. UNIT-IV Data Protection: Confidentiality maintenance of records, Concerns related to data theft in food industry, Role of FDA and FSMA in protection of available food data, Role of ICT in data protection. **Practical work** 1. Maintain a periodic record book of the laboratory work. 2. Risk assessment in your laboratory and report maintenance 3. Study of equipment manuals.
12	**Suggested Resources for Reading** Pierson, M. D., Corlett, D.A., HACCP: Principles and Applications, Springer. Note: Latest edition of text books may be used.